



## 4 Bay RAID External Hard Drive Enclosure for 2.5/3.5 SATA HDD/SSD

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**PRODUCT INSTRUCTION      Model: 804U3RAID3G**



**【Quality Assurance】** Cenmate provides lifetime technical support. If you have any questions about the product, please don't hesitate to contact us, reply and solve your problem within 24 hours.

**Technical Support Contact:**  
**cenmate23@outlook.com**

Hard drives making noise during operation, it is not caused by fans.

We have tested the decibel level of noise with and without the fan---nearly no difference in decibels.

The noise is generated by the hard drive.

# Qusetion

**Q1: After connecting new hard drive to the computer, why can't I find the hard drive and display the disc letter?**

A1: After the new hard drive is connected to the computer, it must be formatted before the system can be recognised and used.

Method of operation: First, right-click 'Computer', then click 'Management', open into the 'Disk Management', find the new hard drive, right-click 'Initialisation 'Select 'MBR (hard drive capacity less than or equal to 2TB)' or 'GPT (greater than 2TB)', then 'New Simple Volume', the default next step to format the partition. Formatting is completed after the hard drive can be used normally, showing the new disc symbols.

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**Q2: The hard drive appears in RAW format in Disk Management due to improper extraction or power failure. What should I do?**

A2: Recovery through the chkdsk command; operation method: administrator identity run cmd input CHKDSK i: /f enter (here the i is the disc letter of the disk), scanning and repair is complete reboot can be displayed.

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**Q3: The hard drive is dynamically invalid in Disk Management since it was previously removed from an old computer?**

A3: This is due to your system. Reinstall the hard drive into the original computer motherboard, and then backed up the data format; If your old computer has been scrapped or can't get into the system, you can download a dynamic disk conversion software to convert directly to a basic disk.

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**Q4: Hard drive just shows up but no disc letter?**

A4: Inside Disc Management, right-click on the Disc Partition section and select Change Disc Path and Name to add a disc name for use.

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**Q5: It automatically disconnects during transmission**

A5: It may be problems of usb interface on computer, usb cable, computer usb driver, hard drive, or hard drive cabinet. You need to contact our technicians to check the problems one by one.

## CONTENT

4 Bay 3.5 Inch Hard Drive RAID Storage  
USB 3.0 A/C Cable,eSATA Cable  
Mounting screws and screw driver  
User Manual  
12V Power Supply

## OS Support

Windows 2000/ Me/XP/  
Vista/ 7/ 8/ 8.1/10 and  
above, Mac OS 9.1/10  
and Linux

## CONNECTING HDD



804U3RAID3G



Into the hard drive and push  
back the hard disk cabinet



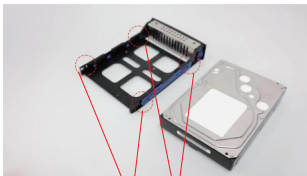
Locate the USB port that  
you will use and connect  
the USB cable.



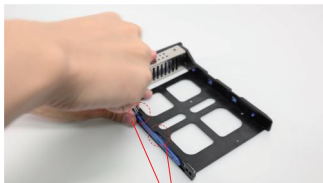
Connect the other side of the  
USB cable in the HDD case.



Connect the DC adaptor  
into the case and case and  
connect the other cable side  
in the wall outlet



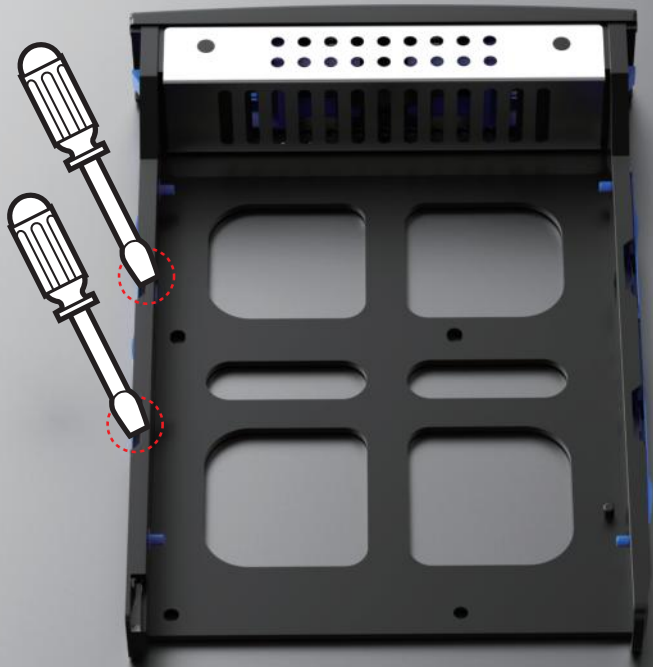
Pull up the blue clasp on both sides of the bracket, put hard disk gently.



For 2.5"SSD/HDD: Use screwdriver open the blue clasp on the side.

# Please Note:

For 2.5"SSD/HDD: Use a screwdriver to pry open the clip on the side and then secure them with screws.



## ! NOTE:

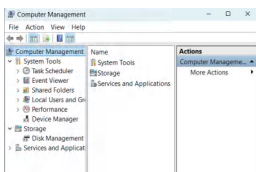
Please follow below information on how to partition and format the drives before use

## ! NOTE:

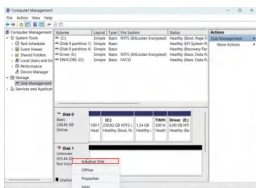
Data will be lost when you reset the unit

## Partition and Format

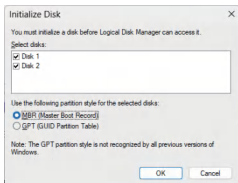
Note: Formatting is only required for new drives



Right click on “My Computer” icon and select “Manager” When this window appears, select “Disk Management”.



Right click on the disk on “New Volume” and select “Initialize Disk”.

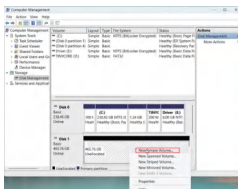


Select the new disk to start with the configuration.

Select:

MBR (hard drive capacity less than or equal to 2TB)

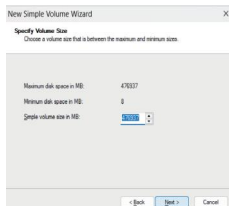
GPT (bigger than 2TB)



When new disk appears, right click on it and select "New Simple Volume"

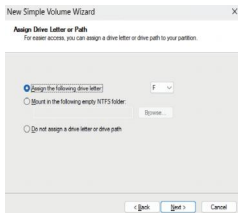


When the wizard window appears, click on [Next].

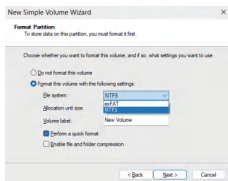


Select the partition size We recommend to use as maximum and click on [Next].

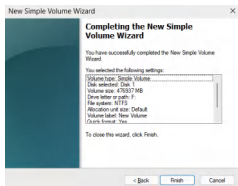




Select the drive letter to sign the hard disk and click on [Next].



Select the format settings: NTFS or "FAT 32", "DEFAULT" and Volume Label (Select the features that you need to format: "FAT32", "Predetermined", the hard drive label that you need (example: "New Volume") and "Perform a quick format".



New partition is complete and your hard disk is ready to be used. Click on [Finish].

# Product Specifications

|                            |                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| RAID Chip                  | JMS567+JMB393                                                                                                                 |
| RAID Level                 | RAID0 RAID1/10 LARGE RAID3 CLONE RAID5 NORMAL                                                                                 |
| Supported Devices          | Supports standard 2.5/3.5 inch SATA I/II HDDs and SSDs                                                                        |
| Cooling Fan                | 2 Inch Cooling Fan                                                                                                            |
| External Interface         | USB 3.0 & eSATA                                                                                                               |
| Maximum Data Transfer Rate | USB 3.0(MAX 5Gbps) / USB 2.0(MAX 480Mbps) / eSATA (MAX 3Gbps)                                                                 |
| OS Support                 | Windows 2000/ Me/ XP /Vista/ 7/ 8 /8.1/10 above, Mac OS 9.1/10. above                                                         |
| Power                      | 12V Power Supply                                                                                                              |
| Plug and Play              | Support                                                                                                                       |
| Package Contents           | 4Bay 3.5 Inch Hard Drive RAID Storage , USB 3.0 Cable, eSATA Cable<br>Mounting screws and driver ,User Manua 12V power supply |

## \*Tool free installation

No need any tools or screws to install for 3.5 inch Hard Drive.

## \*Plug and play

Supports standard 2.5/3.5 inch SATAI/II/III HDDs and SSDs.

## \*High Speed data transfer

Stable data transfer and high speed up to 5Gbps.

## \*Aluminum Case

Provides very good heat dissipation and HDD protection.

## \*Quiet cooling fan technology

Provides very good heat dissipation.

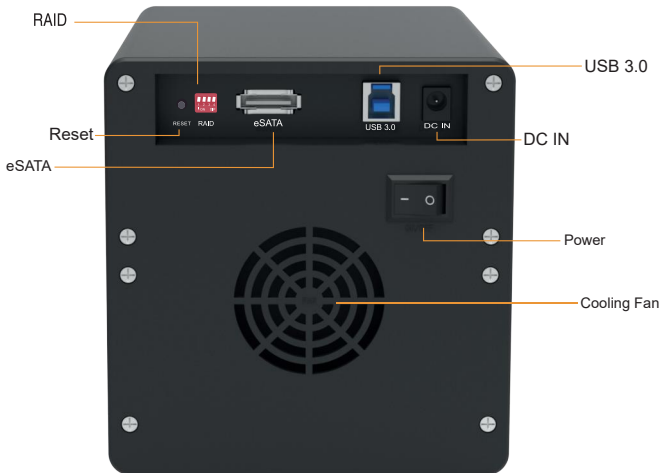
## \*RAID Level

RAID0 RAID1/10 LARGE RAID3 CLONE RAID5 NORMAL.

## \*Hard disk storage capacity

Support up to 22TB on a single hard drive.

# Feature



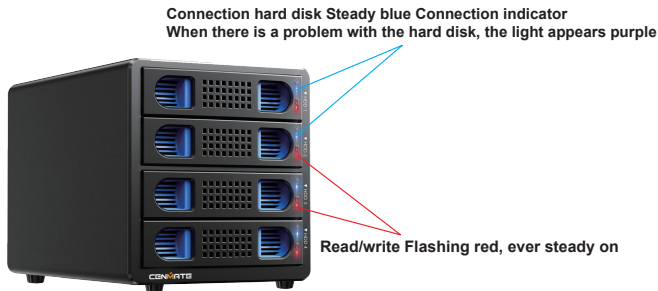
1. RAID Level : RAID 0, RAID 1/10,LARGE RAID3,CLONE,RAID 5,NORMAL
2. USB 3.0 (MAX 5Gbps), USB 2.0 (MAX 480Mbps),eSATA(MAX 3Gbps)
3. DC IN : 12V Power Supply
4. Cooling Fan : 2 Inch Fan
5. Poweron/off: Connect the adapter for power supply, you can use the switch to turn the power on or off.

## USB port change to eSATA port:

It need to turn off the power first, then change the port and turn on the power.If you replace to eSATA interface directly with power on, it will not recognize the hard disk.

# How to use

1. Pull out the tray
2. Install the hard drive
3. Connect esata or USB to the computer
4. Plug in the power supply, turn on the power switch
5. The fan rotates and the hard drive light is on indicating that it is normal and you can start using it.



The connection light is always on blue (indicating a normal connection)  
Read/write light is always red.

Verify the HED indicator If the LED shows purple, the HDD is in an abnormal state. When the blue LED is displayed, the HDD is in normal operation. The HDD reads data and repeats the red LED indicator and blinks during read and write operations

## Hard Drives are Not Recognized

1. Make sure the hard drive has been properly connected and installed (confirm the hard drive can be used normally)
2. Make sure the power supply has been properly connected and energized.
3. Make sure the USB cable is properly connected to the computer.
4. All above connection has done, open the power switch on the back of product, check whether the indicator lights on.

If connected to the computer correctly, the lights are also blinking, then will read the hard drives signal.

**If eSATA only recognizes one drive.**

Is it possible that you are using a computer with an eSATA interface that does not support RAID mode?

There is also a case that the driver you installed only supports recognizing one hard drive. The above two situations will cause only one hard drive to be displayed after connecting the eSATA interface.

## Rebuild RAID10

The time it takes to rebuild the RAID10 mode is based on the size of the harddrive, if the larger your hard drive is, the longer it will take, regardless of whether or not there is data in the hard drive, the above will happen with all RAID enclosures.

When the blue light is always on and the red light keeps blinking, it means that RAID1 or RAID10 is being rebuilt, and when the red light changes from blinking to always on, it means that RAID rebuild is completed.

If you have set up RAID on other brand RAID enclosure, you can't rebuild it on our RAID enclosure.

**Raid Mode Switching Method:** Disconnect the power, use screwdriver toggle the paddle to the corresponding mode, press and hold the reset button, turn on the power, hold reset for ten seconds, the raid mode will be successfully switched.

**NOTE:** When replacing RAID, you need to go back to NORMAL and set the desired RAID mode. For example, change RAID1 to RAID0: You need to change RAID1 to NORMAL mode first, then change NORMAL to RAID0 mode.

MAC OS doesn't have Raid software.

# RAID Level Setting

Be sure to save a good backup of your data before setting up RAID, which will format your hard drive after setting up RAID.



## NORMAL

\*.Each works as a single hard disk, and the data is transferred equally to the mounted hard disk. if four hard disks are installed, the data stored on the four hard disks will be duplicated.



## RAID LEVEL 0

implements a striped disk array, the data is broken down into blocks and each block is written to a separate disk drive.

\*.I/O performance is greatly improved by spreading the I/O load across many channels and drives.

\*.Best performance is achieved when data is striped across multiple controllers with only one drive per controller.

\*.No parity calculation overhead is involved

\*.Very simple design.

\*.Easy to implement (Switch 1 2 3 up & 4 down)



## RAID LEVEL 1

\*.One Write or two Reads possible per mirrored pair.

\*.Twice the Read transaction rate of single disks, same Write transaction rate as single disks

\*.100% redundancy of data means no rebuild is necessary in case of a disk failure, just a copy to the replacement disk.

\*.Transfer rate per block is equal to that of a single disk.

\*.Under certain circumstances, RAID 1 can sustain multiple simultaneous drive failures.

\*.Simplest RAID storage subsystem design (Switch 2 3 up & Switch 1 4 down)



## RAID LEVEL 3

\*.Very high Read data transfer rate. Very high Write data transfer rate.

\*.Disk failure has an insignificant impact on throughput.

\*.Low ratio of ECC (Parity) disks to data disks means high efficiency (Switch 3 up & Switch 1 2 4 down)

## RAID LEVEL 5



Independent Data Disks with Distributed Parity Blocks Characteristics & Advantages

- \*.Highest Read data transaction rate; Medium Write data transaction rate.
- \*.Low ratio of ECC (Parity) disks to data disks means high efficiency.
- \*.Medium Write data transaction rate.
- \*.Good aggregate transfer rate (Switch 2 up & Switch 1 3 4 down)

## RAID LEVEL 10



Very High Reliability Combined with High Performance Characteristics & Advantages

- \*.High I/O rates are achieved by striping RAID 1 segments
- \*.Under certain circumstances, RAID 10 array can sustain multiple simultaneous drive failures
- \*.RAID 10 is implemented as a striped array whose segments are RAID 1 arrays
- \*.RAID 10 has the same fault tolerance as RAID level 1
- \*.RAID 10 has the same overhead for fault-tolerance as mirroring alone (Switch 2 3 up & Switch 1 4 down)

## LARGE(JBOD)



- \*.A non-RAID configuration, known as LARGE (Just a Bunch Of Disks) is a popular method for combining multiple disk drives into a single logical drive.
- \*.LARGE features several hard disk concatenated together to create one logical drive. For example, we can combine a 1TB, 2TB and 4TB drive into a 7TB logical drive. LARGE does not provide any data redundancy (Switch 1 3 up & Switch 2 4 down)



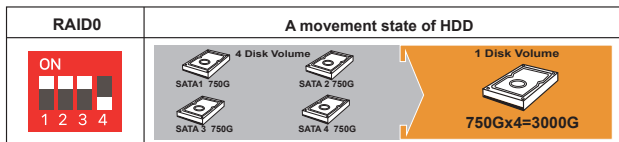
## CLONE

- \*.The operating system recognizes each single disk and uses the hard disk independently.(Switch 1 2 up & Switch 3 4 down)

The DIP switch configurations are the same for RAID1 and RAID10, but when you have four hard disks installed, the toggle DIP switch configuration will be RAID10 mode, and when you have two hard disks installed, the toggle DIP switch configuration will be RAID1 mode.

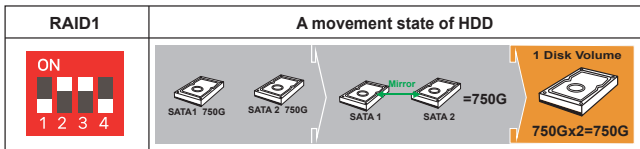


## 1) RAID 0



It consists of two or more hard disks in a striping method and uses all the capacity as one volume. And when recording data continuously, each hard disk alternately records or listens to different data at the same time, so the full speed is very fast. However, if one hard disk fails, the rest of the data stored on the other hard disks will also be lost, so data reliability is poor. RAID0 mode is suitable for temporary data such as real-time rendering data or logs that require high performance.

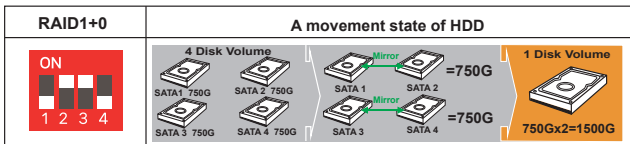
## 1) RAID 1



The mirroring method uses two or more hard disks, each of which stores the same data, so even if one hard disk is damaged, the data can be recovered because there is another hard disk with the same data.

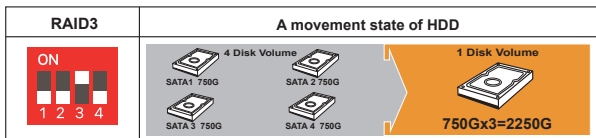
Note-RebuildRAID 1 mode is switched in the same way as RAID 10 mode, so when two hard disks are installed, RAID 1 mode is configured and when four hard disks are installed, it is automatically set to RAID 10. RAID 10 mode supports RAID 0 and RAID 1 at the same time, so you can realize high performance and stability of data backup.- In this mode, please use the "Hot Swap" method to replace the hard disk without turning off the power of the product. When replacing the hard disk due to a defective or malfunctioning hard disk, remove only the defective hard disk and install a new hard disk in its place. If the PC's power is turned off or the USB cable or e-SATA cable connected to the PC is removed during the data recovery process, the data recovery process may be interrupted. If the data recovery process is interrupted, please turn off the power switch of the product and turn it back on to continue the data recovery process.

## 1) RAID 10+(1+0)



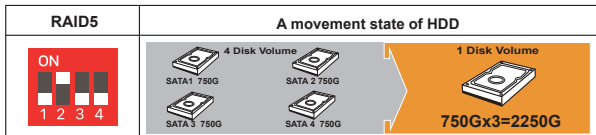
This is a RAID method that meets the superior performance of striping and the high reliability of mirroring by mixing mirroring and striping methods, requiring a minimum of four hard disks. Two sets of two hard disks are configured as RAID 1 and two sets of RAID 1 are configured as RAID 0. RAID 10 Mode: For database or application server data that requires high performance.

## 2) RAID 3



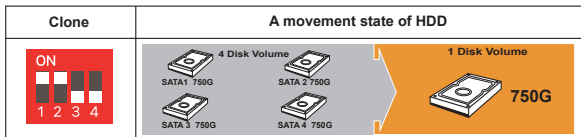
The minimum number of hard disks used is three, and one hard disk serves to store parity information so that if another hard disk fails, the lost data can be retrieved, recovered, and played back. Utilization of RAID 3 mode: Suitable for a single-user system in a business where large records are used a lot. <Note>.record: A unit of data that is used as a unit to actually read and write data when accessing a file.

## 3) RAID 5



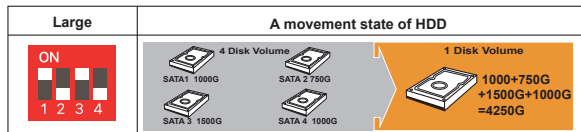
The minimum number of hard disks used is three, and parity information is not stored on any one hard disk, but is distributed across all hard disks so that if one hard disk fails, the lost data can be reconstructed from the data on the remaining working hard disks. Utilization of RAID5 mode: Ideal for databases and web services that do not require high performance.

#### 4) Clone



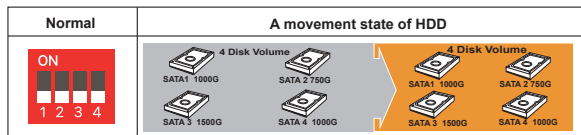
If you have four hard disks installed, the data stored on all four hard disks will be duplicated exactly the same.

#### 5) LARGE(JBOD)



It logically combines multiple hard disks to form a single disk, but does not offer any advantages over using each hard disk independently.

#### 6) NORMAL(Clear RAID)



The operating system recognizes each disk as a single disk and uses the hard disks independently.<References>.Stability of RAID onfiguration :RAID1>RAID 5>RAID 10(1+0)>RAID 0.Transfer speed of RAID configuration :RAIDO>RAID 10 (1+0)>RAID 5>RAID1.

After the RAID, the hard drive is quite initially bought back, need to be initialized, after formatting, the partition can be used normally.

## Caution

Do not connect the 804U3RAID and hard drive (HDD) to a computer when they are not fully assembled. There is a risk of damage and electric shock to the HDD.

This device supports SATA(I/II/III) HDDs! This device is also very teachable to small shocks, so be aware of shocks. Pay attention to the orientation and position of the hard drive during assembly.

Incorrect orientation and position can cause damage to the device.

Since the device may be damaged due to product defects, etc., please use a harddrive(HDD) that does not have important records when checking whether the product is working properly for the first time.

We will not be liable for damages to the HDD used in connection with this equipment. We will not be liable for loss of data recorded on the HDD.

# RAID Setup Software

Need more support, feel free to contact:  
**cenmate23@outlook.com**

**Only for windows**



**[https://www.cenmate.com/index.php  
?m=home&c=Lists&a=ndex&tid=4](https://www.cenmate.com/index.php?m=home&c=Lists&a=ndex&tid=4)**

# Hi

**We are committed to providing consumers with superior products and services.**

**Please contact us if you have any other questions.**

**We really need your help if you are satisfied with our services!!!**

**Could you share your feelings about this product with others on Amazon!?**

**This will help them with their purchases.**

## Thank you

The logo for CENMATE features the word in a bold, sans-serif font. The letters 'CEN' and 'ATE' are in a dark grey color, while the 'M' is a vibrant orange. A small orange leaf-like shape is positioned above the top right of the 'M'.